Case Study A — EDA with initial-phase outlier investigation

Follow-up to `case_study_A_eda.ipynb`

In the first EDA notebook, the within-cycle waveform comparison (Pneumatic pressure / Velocity / Acceleration piston rod, early vs. mid-life vs. near-failure cycle) showed sharp positive/negative spikes at the start of the plotted window, settling down afterwards. This notebook investigates:

1. **What exactly is spiking?** The earlier plot's x-axis is the *within-cycle sample index* (0–99), not the device's life stage — so "settles down after the initial phase" could mean either (a) a transient at the start of *every* cycle's recording, or (b) a genuine *early-in-life* ("running-in") effect across the device's 4,766-cycle lifetime. Part 1 disambiguates the two.
2. **Are there genuine early-life outliers?** Part 2 builds a data-driven detector that flags early cycles whose value differs significantly from the later, steady-state ("a regime") value, per channel.
3. **Does it matter for the EDA conclusions?** Part 3 repeats the key analyses from the first notebook — distributions, degradation-trend correlations, channel correlation matrix — excluding the flagged initial-phase outlier cycles, and compares before/after.

Same prerequisites as the first notebook (`zema_env` kernel, `pandas` / `seaborn` , cached dataset).

In [1]:
```
import sys
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sys.path.insert(0, str(Path.cwd()))
from case_study_A_analysis import Config, load_zema  # reuse the
pipeline's own loader

sns.set_theme(style="whitegrid")
plt.rcParams["figure.dpi"] = 100
pd.set_option("display.precision", 4)

cfg = Config()
X, UX, y = load_zema(cfg)

CHANNEL_NAMES = [
    "Acceleration plain bearing", "Acceleration piston rod", "Acceleration
ball bearing",
    "Active current", "Axial force",
    "Motor current phase 1", "Motor current phase 2", "Motor current phase
3",
    "Pneumatic pressure", "Sound pressure", "Velocity",
]
D = cfg.datapoints_per_cycle
N = X.shape[0]


def channel_block(name):
    i = CHANNEL_NAMES.index(name)
    return X[:, i * D:(i + 1) * D]


print(f"X {X.shape}  |  {N} cycles  |  {D} within-cycle samples/channel")
```

```
X (4766, 1100)  |  4766 cycles  |  100 within-cycle samples/channel
```

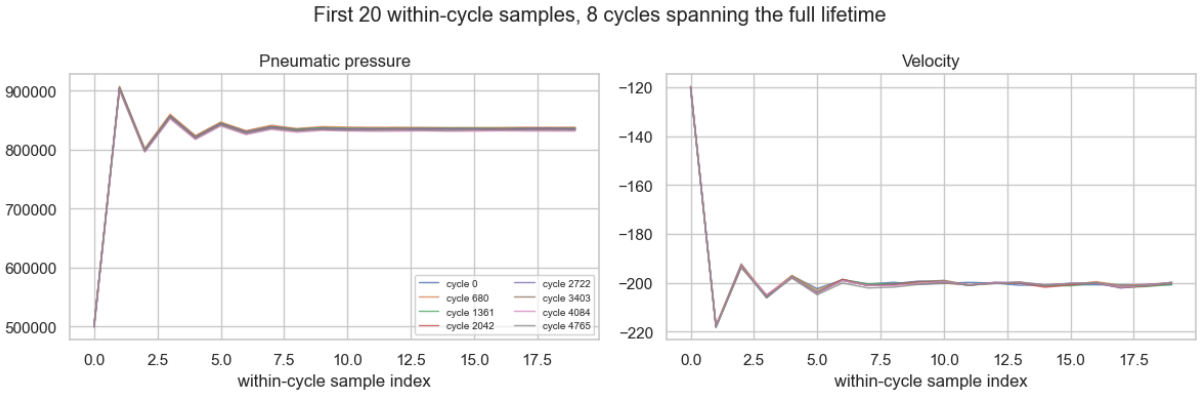
```
/Users/emanuelebuchicchio/repos/case_study_a/zema_env/lib/python3.10/site-
packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update
jupyter and ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

Part 1 — What is actually spiking?

Re-examine the original comparison, but overlay several example cycles spread across the *whole* life (not just 3), zoomed on the first samples of each cycle. If the spike shape/magnitude is the same regardless of which life-stage the cycle comes from, it is a **within-cycle acquisition transient**, not an early-life effect.

In [2]:
```
example_idx = np.linspace(0, N - 1, 8, dtype=int)
plot_channels = ["Pneumatic pressure", "Velocity"]

fig, axes = plt.subplots(1, len(plot_channels), figsize=(12, 4))
for ax, name in zip(axes, plot_channels):
    block = channel_block(name)
    for k in example_idx:
        ax.plot(block[k, :20], lw=1, alpha=0.8, label=f"cycle {k}")
    ax.set_title(name)
    ax.set_xlabel("within-cycle sample index")
axes[0].legend(fontsize=7, ncol=2)
fig.suptitle("First 20 within-cycle samples, 8 cycles spanning the full lifetime")
plt.tight_layout()
plt.show()
```



Every example cycle — early, mid-life, and near-failure alike — shows the same spike at the very start of its recording, then settles to the same steady band. This is a transient at the start of *each cycle's own 100-sample window*, not something tied to the device's degradation stage.

Quantify it: for every channel, compute each sample position's average absolute deviation from that cycle's own median, averaged over **all 4,766 cycles** (so early/mid/late life all contribute equally). A position-dependent spike that survives this averaging confirms a systematic, life-stage -independent transient. The settling length `K` is the number of leading positions whose average deviation exceeds 5× the steady-state baseline level (median of positions 20+).

In [3]:
```
BASELINE_FROM = 20
SPIKE_RATIO = 5.0

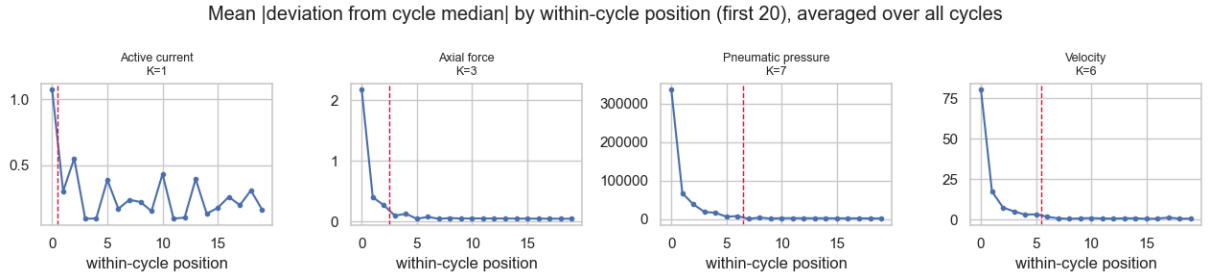
settle_K, pos_dev = {}, {}
for name in CHANNEL_NAMES:
    block = channel_block(name)
    dev = np.abs(block - np.median(block, axis=1, keepdims=True))
    pos_mean_dev = dev.mean(axis=0)
    baseline = np.median(pos_mean_dev[BASELINE_FROM:])
    K = 0
    for p in range(len(pos_mean_dev)):
        if pos_mean_dev[p] > SPIKE_RATIO * baseline:
            K = p + 1
        else:
            break
    settle_K[name] = K
    pos_dev[name] = pos_mean_dev

settle_df = pd.DataFrame({
    "settling length K": settle_K,
    "peak deviation (pos 0)": {n: pos_dev[n][0] for n in CHANNEL_NAMES},
    "steady baseline": {n: np.median(pos_dev[n][BASELINE_FROM:]) for n in
CHANNEL_NAMES},
})
settle_df["peak / baseline"] = settle_df["peak deviation (pos 0)"] /
settle_df["steady baseline"]
settle_df.sort_values("settling length K", ascending=False)
```

Out[3]:

	settling length K	peak deviation (pos 0)	steady baseline	peak / baseline
Pneumatic pressure	7	336042.5765	1137.0793	295.5313
Velocity	6	80.5350	0.5479	146.9947
Axial force	3	2.1724	0.0444	48.9423
Active current	1	1.0744	0.2123	5.0609
Acceleration plain bearing	0	1.3502	1.0974	1.2304
Acceleration piston rod	0	2.7609	4.0021	0.6899
Acceleration ball bearing	0	2.6699	3.1381	0.8508
Motor current phase 1	0	0.7756	2.8360	0.2735
Motor current phase 2	0	1.4924	2.8246	0.5284
Motor current phase 3	0	2.2501	2.6943	0.8351
Sound pressure	0	0.3969	0.4435	0.8950

In [4]:
```
transient_channels = [n for n, k in settle_K.items() if k > 0]
fig, axes = plt.subplots(1, len(transient_channels), figsize=(3.2 *
len(transient_channels), 3))
if len(transient_channels) == 1:
    axes = [axes]
for ax, name in zip(axes, transient_channels):
    ax.plot(pos_dev[name][:20], marker="o", ms=3)
    ax.axvline(settle_K[name] - 0.5, color="crimson", ls="--", lw=1)
    ax.set_title(f"{name}\nK={settle_K[name]}", fontsize=9)
    ax.set_xlabel("within-cycle position")
fig.suptitle("Mean |deviation from cycle median| by within-cycle position
(first 20), averaged over all cycles")
plt.tight_layout()
plt.show()
```



Part 1 conclusion. Pneumatic pressure (K≈7), Velocity (K≈6) and Axial force (K≈3) have a short, systematic settling transient at the start of *every* cycle's recording — present with the same magnitude whether the cycle is from early, mid, or near-failure life. This is far more consistent with a **measurement/acquisition artifact** (sensor or DAQ settling time right after each cycle trigger) than with any physical life-stage effect, and it answers the original question directly for pressure and velocity: yes, that spike looks like a measurement-system issue, not a degradation signal.

One practical consequence: `case_study_A_analysis.py`'s `load_zema()` takes the **first** `datapoints_per_cycle` (100 of 2,000) raw samples per cycle (see `ZeMASamples._extract_data`), so for Pneumatic pressure this transient occupies ~7% of the raw features the model actually trains on for that channel.

Part 2 — Genuine early-in-life ("running-in") outliers

Having isolated the within-cycle artifact, check for a *different* phenomenon: cycles in the first part of the device's life whose value is genuinely out of line with the later, steady-state ("a regime") value — e.g. mechanical run-in/break-in. Method, per channel:

1. Use the **clean** per-cycle mean — the within-cycle mean computed *after* dropping that channel's own settling window `K` from Part 1, so the acquisition transient can't masquerade as an early-life effect.
2. Build a robust steady-state reference (median/MAD) from a mid-life window (cycles 1000–3000), away from both the start and the near-failure end.
3. Flag a cycle in the early window (first 500 cycles) as an outlier if its clean mean's robust z-score exceeds 3 in absolute value.

In [5]:

```
REF_LO, REF_HI = 1000, 3000
EARLY_N = 500
Z_THRESH = 3.0


def clean_mean(name):
    block = channel_block(name)
    K = settle_K[name]
    return block[:, K:].mean(axis=1) if K > 0 else block.mean(axis=1)


raw_flag_counts, clean_flag_counts, masks = {}, {}, {}
for name in CHANNEL_NAMES:
    raw_m = channel_block(name).mean(axis=1)
    cln_m = clean_mean(name)
    for m, store in [(raw_m, raw_flag_counts), (cln_m,
clean_flag_counts)]:
        ref = m[REF_LO:REF_HI]
        med = np.median(ref)
        mad = np.median(np.abs(ref - med)) * 1.4826 + 1e-15
        z_early = (m[:EARLY_N] - med) / mad
        store[name] = int((np.abs(z_early) > Z_THRESH).sum())
    ref = cln_m[REF_LO:REF_HI]
    med, mad = np.median(ref), np.median(np.abs(ref - np.median(ref))) *
1.4826 + 1e-15
    flagged = np.zeros(N, dtype=bool)
    flagged[:EARLY_N] = np.abs((cln_m[:EARLY_N] - med) / mad) > Z_THRESH
    masks[name] = flagged

outlier_df = pd.DataFrame({
    "settling K": settle_K,
    "n flagged (raw mean)": raw_flag_counts,
    "n flagged (clean mean)": clean_flag_counts,
})
outlier_df["frac flagged (clean, of first %d)" % EARLY_N] = outlier_df["n
flagged (clean mean)"] / EARLY_N
outlier_df.sort_values("frac flagged (clean, of first %d)" % EARLY_N,
ascending=False)
```

Out[5]:

	settling K	n flagged (raw mean)	n flagged (clean mean)	frac flagged (clean, of first 500)
Axial force	3	374	375	0.750
Motor current phase 1	0	346	346	0.692
Motor current phase 2	0	64	64	0.128
Acceleration ball bearing	0	52	52	0.104
Velocity	6	172	44	0.088
Active current	1	31	27	0.054
Sound pressure	0	11	11	0.022
Acceleration plain bearing	0	9	9	0.018
Pneumatic pressure	7	1	1	0.002
Acceleration piston rod	0	0	0	0.000
Motor current phase 3	0	0	0	0.000

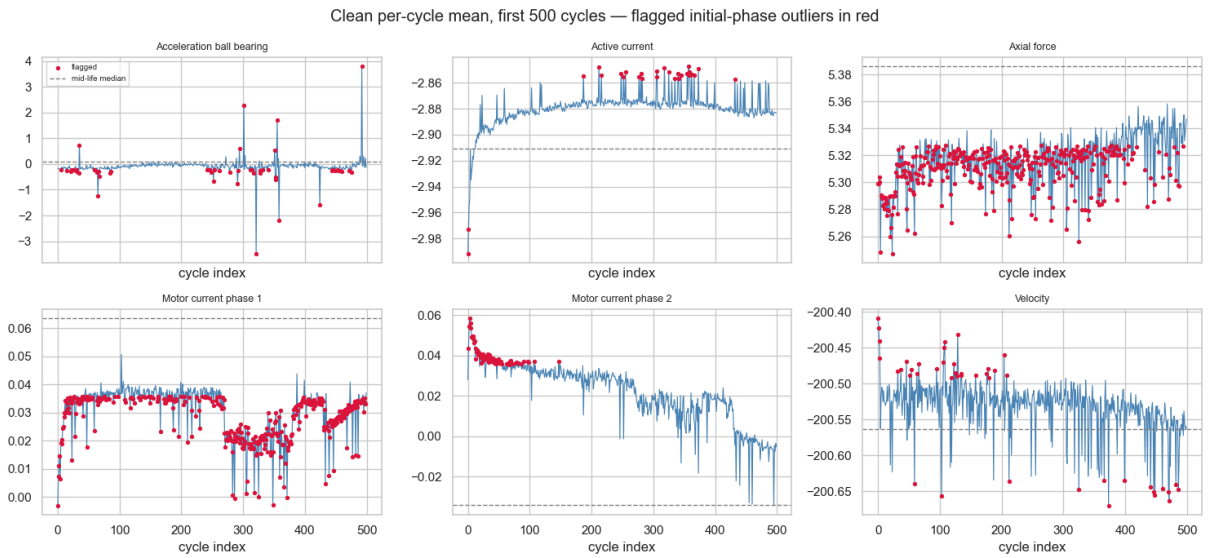
For Pneumatic pressure, the clean-mean flag count stays at essentially zero — confirming Part 1: once the acquisition transient is removed, pressure shows no real early-life effect. Velocity and Active current drop sharply from the raw-mean count once the transient is removed, but keep a smaller residual. Axial force and Motor current phase 1 stay large either way (their settling window `K` is 0–3, too short to explain hundreds of flagged cycles) — a genuine, extended early-life regime shift, consistent with mechanical run-in/break-in.

Channels are marked as having a genuine initial-phase effect if more than 5% of the early window is flagged on the clean mean.

In [6]:
```
QUALIFY_FRAC = 0.05
qualifying = [
    n for n in CHANNEL_NAMES
    if outlier_df.loc[n, "n flagged (clean mean)"] / EARLY_N >
QUALIFY_FRAC
]
print("Channels with a genuine initial-phase effect (>%.0f%% of first %d
cycles flagged):"
      % (QUALIFY_FRAC * 100, EARLY_N))
for n in qualifying:
    print(f"  - {n}  ({outlier_df.loc[n, 'n flagged (clean
mean)']}/{EARLY_N} cycles)")
```

```
Channels with a genuine initial-phase effect (>5% of first 500 cycles
flagged):
  - Acceleration ball bearing  (52/500 cycles)
  - Active current  (27/500 cycles)
  - Axial force  (375/500 cycles)
  - Motor current phase 1  (346/500 cycles)
  - Motor current phase 2  (64/500 cycles)
  - Velocity  (44/500 cycles)
```

In [7]:
```
fig, axes = plt.subplots(2, 3, figsize=(15, 7), sharex=True)
for ax, name in zip(axes.flat, qualifying):
    cln_m = clean_mean(name)
    flagged = masks[name]
    cyc = np.arange(EARLY_N)
    ax.plot(cyc, cln_m[:EARLY_N], lw=0.8, color="steelblue")
    ax.scatter(cyc[flagged[:EARLY_N]], cln_m[:EARLY_N][flagged[:EARLY_N]],
               color="crimson", s=8, zorder=3, label="flagged")
    ref = cln_m[REF_LO:REF_HI]
    ax.axhline(np.median(ref), color="gray", ls="--", lw=1, label="mid-
life median")
    ax.set_title(name, fontsize=9)
    ax.set_xlabel("cycle index")
for ax in axes.flat[len(qualifying):]:
    ax.axis("off")
axes.flat[0].legend(fontsize=7)
fig.suptitle("Clean per-cycle mean, first %d cycles — flagged initial-
phase outliers in red" % EARLY_N)
plt.tight_layout()
plt.show()
```



Part 3 — Repeating the EDA excluding initial-phase outliers

Build one combined outlier-cycle set from the qualifying channels (majority vote: flagged by at least 2 of them, to avoid one noisy channel driving the exclusion on its own), then redo the distribution, degradation-trend and correlation views from the first notebook with vs. without those cycles.

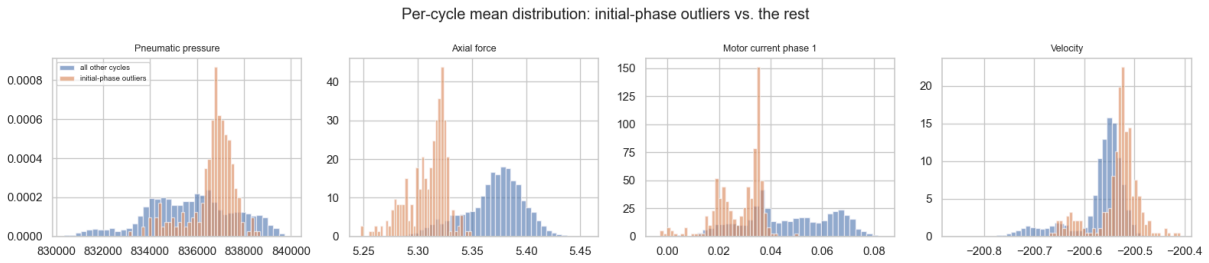
In [8]:
```
VOTE_THRESH = 2
vote = np.sum([masks[n] for n in qualifying], axis=0)
is_outlier = vote >= VOTE_THRESH

print(f"Combined initial-phase outliers: {is_outlier.sum()} / {N} cycles "
      f"({is_outlier.sum() / N:.1%}), spanning cycle index "
      f"{np.where(is_outlier)[0].min()}–{np.where(is_outlier)[0].max()}")

df_mean_clean = pd.DataFrame({name: clean_mean(name) for name in
CHANNEL_NAMES})
df_mean_clean["is_outlier"] = is_outlier
df_mean_clean["y"] = y
```

```
Combined initial-phase outliers: 286 / 4766 cycles (6.0%), spanning cycle
index 0–493
```

In [9]:
```
show_channels = ["Pneumatic pressure", "Axial force", "Motor current phase
1", "Velocity"]
fig, axes = plt.subplots(1, len(show_channels), figsize=(16, 3.5))
for ax, name in zip(axes, show_channels):
    ax.hist(df_mean_clean.loc[~is_outlier, name], bins=40, alpha=0.6,
label="all other cycles", density=True)
    ax.hist(df_mean_clean.loc[is_outlier, name], bins=40, alpha=0.6,
label="initial-phase outliers", density=True)
    ax.set_title(name, fontsize=9)
axes[0].legend(fontsize=7)
fig.suptitle("Per-cycle mean distribution: initial-phase outliers vs. the
rest")
plt.tight_layout()
plt.show()
```

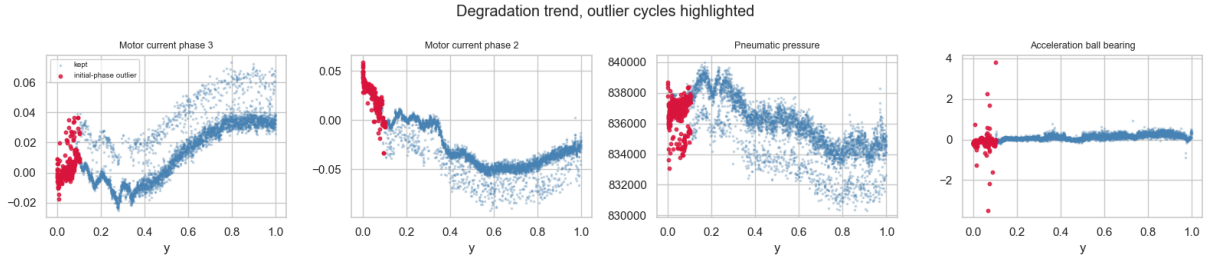


In [10]:
```
corr_all = df_mean_clean[CHANNEL_NAMES].corrwith(df_mean_clean["y"])
corr_excl = df_mean_clean.loc[~is_outlier,
CHANNEL_NAMES].corrwith(df_mean_clean.loc[~is_outlier, "y"])
cmp = pd.DataFrame({"corr with y (all cycles)": corr_all, "corr with y
(outliers excluded)": corr_excl})
cmp["|delta|"] = (cmp.iloc[:, 0] - cmp.iloc[:, 1]).abs()
cmp.sort_values("|delta|", ascending=False)
```

Out[10]:

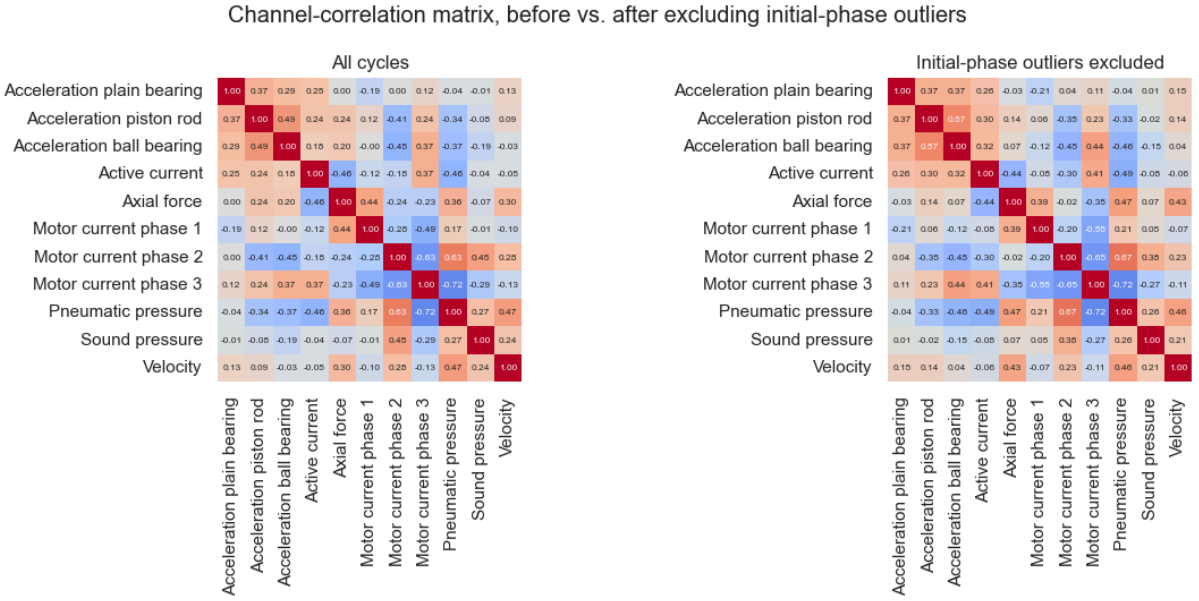
	corr with y (all cycles)	corr with y (outliers excluded)	|delta|
Axial force	0.1833	-0.0050	0.1883
Motor current phase 1	-0.1801	-0.3160	0.1358
Active current	0.1167	0.2099	0.0932
Sound pressure	-0.3014	-0.2227	0.0787
Velocity	-0.0990	-0.0388	0.0601
Acceleration ball bearing	0.5407	0.5980	0.0574
Motor current phase 2	-0.7387	-0.6862	0.0525
Acceleration piston rod	0.5152	0.4748	0.0404
Pneumatic pressure	-0.7289	-0.7566	0.0277
Motor current phase 3	0.7458	0.7644	0.0187
Acceleration plain bearing	0.1200	0.1110	0.0090

In [11]:
```
top = corr_all.abs().sort_values(ascending=False).index[:4]
fig, axes = plt.subplots(1, len(top), figsize=(16, 3.5))
for ax, name in zip(axes, top):
    ax.scatter(df_mean_clean.loc[~is_outlier, "y"],
df_mean_clean.loc[~is_outlier, name],
               s=2, alpha=0.25, color="steelblue", label="kept")
    ax.scatter(df_mean_clean.loc[is_outlier, "y"],
df_mean_clean.loc[is_outlier, name],
               s=10, alpha=0.8, color="crimson", label="initial-phase
outlier")
    ax.set_title(name, fontsize=9)
    ax.set_xlabel("y")
axes[0].legend(fontsize=7)
fig.suptitle("Degradation trend, outlier cycles highlighted")
plt.tight_layout()
plt.show()
```



In [12]:

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5.5))
sns.heatmap(df_mean_clean[CHANNEL_NAMES].corr(), annot=True, fmt=".2f",
cmap="coolwarm",
            center=0, square=True, ax=axes[0], annot_kws={"fontsize": 6},
cbar=False)
axes[0].set_title("All cycles")
sns.heatmap(df_mean_clean.loc[~is_outlier, CHANNEL_NAMES].corr(),
annot=True, fmt=".2f", cmap="coolwarm",
            center=0, square=True, ax=axes[1], annot_kws={"fontsize": 6},
cbar=False)
axes[1].set_title("Initial-phase outliers excluded")
fig.suptitle("Channel-correlation matrix, before vs. after excluding
initial-phase outliers")
plt.tight_layout()
plt.show()
```



Summary

- **Pneumatic pressure / Velocity "spikes":** these are a **within-cycle acquisition transient** (the first ~6–7 of 100 samples in *every* cycle, at every life stage), not an early-device-life effect. It looks like a measurement/DAQ settling artifact rather than a degradation signal — for pressure specifically, once it's accounted for there is essentially no residual early-life deviation from the steady-state value.
- **Genuine initial-phase (running-in) outliers do exist**, but on different channels: Axial force and Motor current phase 1 show an extended early-life regime shift (most of the first few hundred cycles deviate strongly from the mid-life reference) consistent with mechanical run-in/break-in; Velocity, Motor current phase 2, Acceleration ball bearing and Active current show a smaller residual effect after the acquisition transient is removed.
- **Effect on the EDA conclusions:** excluding the combined initial-phase-outlier cycle set shifts the degradation-trend correlations by the amounts in the table in Part 3 (see `|delta|` column) — the ranking of which channels track consumed-lifetime fraction most strongly is essentially unchanged, but the magnitude of a few correlations moves measurably once the run-in cycles are excluded.
- **Actionable takeaway:** this is a data-quality finding upstream of `case_study_A_analysis.py`, not a change already reflected in `case_study_A_report.md`. Two independent, separately addressable issues: (1) trimming the first few within-cycle samples for pressure/velocity/axial force before feature extraction would remove a likely acquisition artifact from the model's raw inputs; (2) treating the first ~300–500 cycles as a distinct "run-in" regime (e.g., excluding them from training, or modeling them separately) would remove a genuine but non-representative mechanical transient from the degradation model, if confirmed against the original ZeMA documentation/metadata.